

Video script - HTA in Europe

And that you have. It's animation. Round number 4, I think it looks like on our training session adventure. So this one is more on HT in Europe and key insights into major markets and implications for global teams. And I think from, from my side, you know, some of the rationale as to why we wanted to talk about this, or at least, you know, hear a little bit more on this, was some of the conversations we've been having, even with our Gen X, is like, uh, you know, insight into differences between the markets and it could be from an HTA standpoint or from a bear standpoint and how, you know, people who have global. Well, responsibilities need to deal with that or handle that. And obviously, each company, I'm sure, is a little bit different, but obviously there's an overarching blank and consistency to them. So, some of my thinking at least in the schedule is coloured by the recent experiences with, you know, groups like Argenics, but, um, at least for the conversation today, hopefully it's informative on HT in Europe. And we can, I know, as Paul is chair, previously, okay, there's some variations and differences with Canada and Japan and others, but I think it should provide a pretty strong rationale for, um, understanding, uh, major market pairs and HTA perspectives. Right. Restart. Yep. Sounds okay. The 1st light is blank. Space here. This is... Yeah, sorry to interrupt. I just saw I enjoyed it. I Hello, hello. We just started the introduction, Simon. Yeah. Yeah, yeah. Okay, so, I was just saying that you have a slide here shows... Yeah, we have nothing on. This was exactly what we encountered in 1978, in Europe, when there were discussions about HTA in the UK or in England, uh, more specifically, we're starting. Everybody was interested in it, but nobody knew what was coming inside the review cycle, or what impact this could give on marketing new products. From a European perspective, And now we will talk primarily about European perspective. HTA was something which was considered to be exotic. It was a pet project of the Australians, and some of the Canadian review boards, yeah. And the majority in Europe believed that this would not fly in Europe at all. So they were actually starting. We have a strategy to stop that implementation, or at least to consider this a push fire where you could build a wall around and say, okay, don't further. Let's limit it to England, end over. And since the late '90s, the bushfire has spread, spread all over the place in Europe, although not everybody will accept that, that this has spread, but HTA is

everywhere, in Europe, but it has different meanings, and I have different impacts on the decision process. So if you move to the next slide, you will see, and this is actually one of the slides that be used now for more than 15 years, yeah, in terms of trying to explain, what an HTA review really wants to show. And we have to see on the left hand side, cost effectiveness. So the first starting position of nice, nice as an institute was inaugurated, late 90s, and it was called National Institute of Cost Effectiveness. Only 10 years afterwards, they changed the name in National Institute of Clerical excellence, yeah? But the starting position was that you could, let's say, construct a mathematical process where you are calculating, cost effectiveness, and based upon cost effectiveness, you would decide, a product can be used in the healthcare purchased, or a product cannot be funded by healthcare budget. And it was a yes or no, you had no gray zone in between, yes or no. So that is cost effectiveness. This was very much what the Australians were using. Yeah, they were very strong, very dogmatic in terms of their cost effective ratios. Canada also, from that perspective, but on continental Europe, only a few smaller countries were following that lead. One of the countries was Sweden. where cost effectiveness was spearheaded, but was not used in the decision on the price, yeah? It was used more from an academic perspective, but in the price decision. In other countries, like Holland, they also want to be very close to the English environment, but appear about it by itself. They were saying, yes, I take your argument or cost effectiveness, but I do not use this data to decide on funding. Yeah, I want to see the cost effectiveness, but there's a dissociation between seeing the data and deciding on cost effectiveness ratios. In the UK, it was always clear, if you are below the threshold, then you are cost effective, then you are funded 100%, and again, other countries work in the middle, and there were some very notorious countries, which were completely against cost effectiveness, calculations, and HTA, decision process, which was based across effectiveness. and make a jump immediately into the blue, white, and red, button there, which is the French button. The French consider themselves as the inventors of HDA. And they already have a process from the '80s, where they were forming two review bodies. One review body was a transparency committee, and the other review body was an economic committee, and they're working independently from each other. So which means that everything that is coming from a regulatory process. This was has to be reviewed, and pronounced by a transparency committee, where they're looking at the clerical excellence, they were looking at the data, and based upon a formula and a rating that you get.

These data are brought to an economic process where prices being negotiated. So, from a clear perspective, priced is not part of the equation. Probably a normal perspective, they only are working on the value that has been given to them. It's not on cost effectiveness whatsoever. Only in, let's say, 10, 15 years ago, they started hesitantly to also demand a cost effectiveness model. If your aim is, you're going for a true innovation, they would like to see cost effectiveness data, a model, which is, Let's say, not so defined, well defined as the English cost effect in this model, used by nice, is more latitude in design, but they want to see the results, and they're still want to decide for themselves. We are using this data as an input in our decision process, or we are not using this data. Yeah, so that's almost an opt in opt out process that they have. The country in the middle is a country that is absolutely not interested in qualities, and cost effectiveness, and that is Germany. And Germany, historically, it was a free pricing market. You could introduce your product at your price, your list price, and you had to defend the value of your product, to the individual prescriber, and the sickness funds they were funding this product. The other sickness funds they were using, secondary measures to control the budget. Yeah, so doctors got a target, how much, uh, their, uh, the drug budget could be on an annual basis, if they, if you were overshooting that as an individual, physician, specialist. You always had to pay back one by one for one to the sickness one, the crown, the overspent that you did. So the individual treating position was also very cost conscious, yeah? I mean, he will not use very expensive treatment. If there was an alternative treatment, which was significantly cheaper, Yeah, and that he considered to be almost of equal value, not the same value of equal value. So, as this system worked for 10, 15 years, satisfactory, budgets were, let's say, exploding in Germany, and then they settled a new system, in 2010, the unknown system, where you actually had to prove added benefit over an appropriate competitor. And each word has a senior good meaning, because added benefit means that there would not pay for an equal benefit. I mean, if it was equal benefit or minimal added benefit, then you were talking about a new product being made available at a genetic price level. Yeah? And the problem with the competitive, a comparative, could be a very expensive biotech product, cost 10 to 100,000 a year, or it could be a cheap \$20 per month product, yeah, and the whole discussion with the system where I'm know, was what is the appropriate comparative, what is being used by German physicians, to treat these patients, what is your appropriate compared to what is the added benefit? So we have three different systems. And

then, of course, you have the US, you're not going to discuss the US, but there, affordability also plays a role, to a certain extent, in the discussion. Here, what we see here has very little to do with containing the costs of drugs. It's by setting an individual price of the elevation, and the innovation. By these measures, actually, would allow them later on, the government to make priorities. I'm going to have a product of a high differentiation from the competitor. If I'm allowing this to the market, to the patients, I'm going to dictate that only this project can be used in the disco conditions in this patient group for so long, and only in this stage of the disease, yeah, after you have to write one, two, or three other initial treatment. So, a treatment paradigm is very important, A, uh, identification of the individual patient that is eligible for three industrial important, and all these elements in Germany and France, are coming from the clinical trials. These are the champions of doing a clinical trial analysis. They are the French, for instance, they are notorious of only looking at the phase three trials, and they are looking at the phase three trials individually. You cannot come to them with an integrated database analysis. Not with a better analysis, even if this is of cocrain, evidence, yeah, level, they would consider what is the clear, like, a data set that you have submitted to the French authorities, originally allowed to aim up, and they will analyze this clinical data again. So they do not trust the EMA analysis, or even the French analysis. They do it, again, they analyzing again, to see that the claims that you're making, even if they would be approved by the regulatory authorities, per CHMP, would also be of value for the French patients. So they are a part of the CHampion in Europe, so they are vote for regulatory approval, yes, or no. So the French have a voice there, but they say, okay, we are giving the European patient access from a clinical perspective, but even with our green line, we are still going to review the clinical data. And reviewing the clinical data, from French perspective, is very Cartesian, yeah? I mean, the character was a French mathematician, yeah? He was very strict. And they say, Okay, your clinical trials. What is the patient population of you included in your recruitment, yeah? Is this a reflection of European, or more, specifically, a French patient population? So if you have clean the controls, and this was, let's say, 10, 15 years ago, an issue, if you clean the controls, I'm coming primarily from Russian patients, yeah? And this Russian patient, where most of the time, also Russian patient from Siberia, hospitals, where it was very cheap and fast to get recruitment, and data said, They were extremely suspicious. And then they pleaded towards the European authorities from Ava perspective, to include the true European population, so that

they would reflect that target population, and they would even prefer, they could not demount that to Ava, but for their local review, they were preferring to have a sufficient high number of French patients in the course, where they could do an even a subanalysis to see whether the outcome of the clinic was consistent with the general population in the French group, just to illustrate how important it is, to have not only global data, and global evidence, but sometimes for this type of HD abuse, that they want to see also what is in the books in a local perspective. Their data set, and analysis is very strict. If the marketing team was willing to make a marketing claim in terms of treatment effect, This could not be used in France. St. France, you first have to look at your clinical data, what are the recruitment data of the patients? What are your exclusion criteria, to exclude patients, yeah? So that the French knew very well, whether you were talking about a reflection of the real French society, or that you have an rich data set, yeah, where you're only overtaking the so called patient that would respond best to your treatment, yeah? So that you are cutting off the lightest patient group that will not respond and not show statistically significant benefit for the worst patients that would never be able to respond to your treatment group. So they were seeing whether you were doing some... So, dating ascent, enrichment, to alt up your clinical trial and clinical effect. So the French are extremely rigourous of that. They're also rigourous on whether your clinical endpoints are pre-specified, and are alpha protected. So, they are not willing to hear that you have in your data set, is a primary endpoint, a secondary endpoint, and that this secondary endpoint is exploratory, and that you're willing to make a clay, or the exploratory endpoint, they will wipe it off. Yeah? They will say, no, we are not willing to consider that. If you have a secondary endpoint, and it is alpha protected, we are taking it with us in our analysis, but only as for them as supporting evidence, if your primary outcome has been extremely robust. For clinical teams, this is very, very, very difficult to accept, from a global perspective, yeah? Because this is much more rigourous, then even Emma is willing to do what FDA is willing to accept. Yeah. So, they are more strict to that, and they consider this HDA reviews. If you are in Tubridge here, yeah, and you're going to give it a clinical people, and the rectory people, they're going to see this, this is just an HDA. But the people at the HGV review are, let's say, the top professors in the country, both in Germany and in France. In Emma, they are sending the junior professors, yeah? So these French professors, with their bull dies, yeah. They're considered themselves not as gold, but as the sun king, yeah, the we got ours themselves, in their practice, in their specialty. So

you will have to go with extremely strong data to convince them on certain points. That's why it's so important to go to the French review with a very good plan, to collect the right data, and also prepare an extremely strong transparency file. Yeah, and the quality level that clinical added value data set that you need to prepare, is at least of the same level as the nice file, but most of the time is significantly higher, because you are doing more data set analysis in order to support your primary claims. So that's from a French perspective. The French are giving you ratings, yeah? They will say, This is an ASA bar. One, two, three, four, five. By definition, their mentality is, I give you five. No benefit over anything else in the market that is being used, yeah? And without that, this product should be reimbursed. That's the attitude. The law is not written this way, but this has been the attitude. If you are going off with positive, two positive, clinical trial data. Then you have an opportunity to go for a four. But the point in France is that they are only willing to give you a better rating, if you have data to show that you are having a significant better benefit over a competitor. And the French are very clear, we are not liking side by side comparisons. We don't like cocaine analysis, with hazard ratios and so forth. We would like to see a little control, prospectively, yeah, where you are comparing with a competitive. So we've got super control trials that are just good for FDA, or non inferiority trials, against a active competitor that is sufficient for Emma, Brune, most of the time, are certain optimal to get a clinical added value in France. They are looking for comparative data, but you prove prospectively that you are superior. The exception is that if you're coming within an environment, where actually nothing works, yeah? And that is most of the time when you're in the treatment paradigm, when you have first line treatment, but first line treatment, this insufficient, you go to second line, but that is insufficient. You have nothing left. You're coming with the third line, and if you are putting yourself in third line position, then you can claim there's no competitor, because after the second line, nothing is dead. And then you can have a good rating without having prospective clinical trial data. But most of the time, it's a very hard job, and it needs a lot of consulting, especially to American teams. If you're working with American companies, or buying companies to explain to them, that if they want to be successful in Europe, of course, nice is all in English, yeah? They're trying to publish it that they're everywhere, that is poor, there will be hundred nice presentations, yeah, and there will be zero or 0.1% French presentations, yeah. But they have to look very carefully, how can we be successful from a French perspective? If they know the requirements, then they

can direct, they clear like a drug program in such a way, that they already put in their statistical analysis plan, and pre specified analysis, to reflect the needs, for instance, through the French authorities. And if they have to be specified that in the clinical trial, the 12 digits that Emma will pick it up, put it in their assessment, and if it is in the assessment of the regulatory group at EMA, you will bring this data on a golden plate to the French transparency company, say, yeah, we have done all this data, and the review has been done from a European regulatory perspective, as solid data. But, of course, you have to be sure that Emma also approves or gives a sufficient strong support for this analysis. So that's the French environment that you'll be working with. There are other countries which are working in a very comparative way. For instance, the Italian environment, the Spanish environment, yeah, but there are other, let's say, other peculiarities in these countries that are going beyond that. Before I'm moving now to the economic part of the HDA review, let's look down to Germany. The Germans are... the key mathematicians, because they want to calculate the pit value... No, yeah. They are extremely precise, yeah? The French, they are pragmatic sometimes, but they are dogmatic in terms of added value. The Germans, they want to be extremely rigid, when it comes to who is eligible, what has been used in this calculation, why has it been used? So this appropriate comparison, you have to do a lot of these of the German treatment environment, to know, if you have, friends, the treatment of Dushen, what is being used. And if the competitor is, are we using, bread is alone in 90% of the cases, and then we can switch to long landing, injectable, predis, and then they will review a new product in DMD that claims to have a effect in DMD, against the appropriate competitor, which is used in the market. So if we have now a gene therapy, of course, they will look at it. And they are modally having a standard of care. It's a mix of... various steroids, starting with brandisolo that is costing one dollar for 100 years of treatment, yeah. And some cheap, generic product. For Germany, the first thing you have to do is to say that this is not a comparative, because it is used in Germany not to treat disease, but to trees, some discomfort to patient that has nothing to do with the disease. By itself, the product has no problem to stop the progress of the MD has not proven to have a clinical effect when you... not stabilize it, but certainly not modify the disease, so if you are coming with your arguments towards the Germans, this compare the discussion is 95% of your homework that you have to do, yeah? I mean, we can be extremely helpful in prepare all the evidence so that the German affiliates of large companies, or small companies, can confidently go to this engineering view,

by arguing on the appropriate comparative, because the start of the process is the comparative. And you have to be sure that your program is comparative, princess could be actual skipping products, rather than practice own and support. So what we are going to use in the cost effectiveness model for England, Canada, or ice in the US, will not be the same as what you will use in your arguments in Germany. It will be very different, yeah? And the arguments, I mean, it can be in conflict with each other, but the conflict that you are generating is dependable because you are going to discuss the peculiar German situation of treatment, yeah? President England. In England, you will have primary care, secondary care, and specialty care, that does not exist in Germany. In Germany, like in countries like France and Belgium. Spain and Italy, you can decide as a patient to go to your GP, but you equally can design to go to a specialist, or you can go to a specialty clinic in a hospital or an academic hospital. That's your decision. In England, that's not your decision. England, also, in all of us here in Scandinavia. Your GP will decide whether you need treatment. And if you decide you need treatment, and assessment, that he will refer you to a specialist, where you have a wrong waiting list, huh? For months and months to come. And then this specialist can decide, oh, I'm not competent in this field, is to specialize, I will move to a specialty clinic. And that's only in Amsterdam, or in Rotterdam, or in Uppsala and Sweden. So people have to travel 800 kilometres to get through this clinic. So that's the cost effectiveness discussion. That's not in the German discussion. German discussion is that the individual GP may even decide to use an extremely expensive treatment for a patient, yeah? Never in these cost effectiveness countries. That's why these assessments systems are so different from each other. So that's the German perspective. The German perspective has been the nightmare for big companies, yeah? They were coming with price points of \$10,000, 10,000 euro, and the system, I mean, we only can give you 250. Yeah, that's it. That's what you have proven, yeah? I mean, we can discuss. We can include some qualitative statements that you're making. For instance, you're improving the healthcare, uh, caregiver burden, yeah, so we can go for 250 to 275. But that's the order of magnitude, yeah? The weekly group is there, unless you have been able to convince them that there is no comparative, and if there is no comparative, then they will compare it to European price levels, you know? So that's, from a German perspective, an extremely important element here. If you have this HD assessment being done, that starts the price negotiation, because the price negotiation is something that is done by economically trained by the politicians or government

officials, and they know very well that a price leads to a budget and to a budget impact, yeah? So, for instance, in the UK, when you have a cost effectiveness, model being presented, and the evidence review group, and I think it's now called the evidence, external assessment, you know, yeah, they change names, but it's all the same. These evidence review groups, they are, typically, university groups that are funded by nice to do these assessments, yeah? And they are, at my time, there were about six big ones in the UK. Plus you have a separate one in Scotland. And they were competing for the work from Nice. Yeah, sometimes they were competing for the work by being extremely dogmatic and rigorous, yeah? I mean, you had the Aberdeen group, who could be extremely Scottish stuff, and you had the Wessex groups, from Southeast, who could be more, I'm not saying industry minded, but more open to discussion. So even there, with the same set of data, you could have a different view perspective. But these people, they are giving an assessment, and they are heard in a review meeting, with not only clinical people around the table, experts and cost effectiveness, but also experts in care, nurses, patients, politicians, and they would decide, based upon the input that you have, from a cost effectiveness model, from an evidence review group, whether the product should be funded by the National Health Service. And there are elements where you have, here, society perspective, can play a role. But don't count on that. I personally had the experience that we had a... for a buying tech project, we had a very innovative product against prostate cancer. It was a fusion, uh, kind of, uh, treatment where you were making a fusion between the own tea cells and a new product. And it was the first one to prove that you had a effect on overall survival in treatment of third stage of prostate cancer. Other, the other products all had effects on progression free survival, yeah? But we had over surviving benefits, the Holy Grail. But there were some problems in the U.S. when the analysis of the data, and FDA had requested a complete a reanalysis of the data, and an audit of all the centres to see whether the data that were in the process were correct. So there was, after the submission to FDA, and all the internal the data and the reanalysis, and ultimately, the data set was altered, but still, the overall survival helps strong. When we were going to England, After you were proved that we were cost effective, that we were supported by the FDS review group. In the clinical data set, and in the cost effectiveness data, where we were sitting around the table, where the board, the nice review board, of the data. One, and was strongly supported by the clinical expert on this board, the one that was most negative about our data set, but coming from the patient, yeah? Patient

doubted that the data set was strong enough to warrant that patient would be exposed to this new treatment. Despite the fact that this was the first product ever to show overall survival benefit, whether this was whether this was a very targeted, specialized treatment, individualized patient, based upon the comments that are coming from this patient group. The rest of the review said, if the patient do not want to have this type of, you know, the product, why should we give access to this, you know, the product? So we had a very negative feeling in the group, yeah? based upon this review, and this was totally unexpected, yeah? So it's not only cost effectiveness that plays the roles here, it also the view of, from a society perspective, where the patients want to have, let's say, this type of new innovations, yeah? And the reason why this patient group sometimes are difficult is, it's a very English approach. They want to stand above the ground, and they want to say that they are defending the good of their temptation group, against, let's say, the pressure of the pharmaceutical accomplish that they are not considered as an ally, but as an enemy, because they are taking away or circling away resources in healthcare to very expensive treatments, yeah, taking away that from normal care. So this is something that you have to take into consideration. HDA is not a mathematical review, but it's something where a process needs to be followed very carefully. What can IMS do, actually, in this process, is actually prepare the ground for, if you do an early assessment for a young, upcoming buying that company to explain very carefully that they have to understand that a too small patient pool, yeah, with a, let's say, a very specialized endpoint in the clinical trust, which is not considered to be patient relevant, that this will not fly in the majority of the countries, yeah? For instance, in Germany, you have to have endpoints of your clinical trial, which are patient relevant. What is that, me? I mean, if you are lame, and you're giving you treatment, then your patient has to walk again. So what are you doing? You're measuring, how many patients are work again, in a very way. But you are not saying, having a scale, yeah, where, I'm testing, what is the muscle, after, before treatment, then during treatment, and after treatment, every see a significant increase of this muscle. So we have met the endpoint, it has a P value of no point, no, no, no. One, the Germans will say, show me the relevance, yeah? Beans, have you been able to prove that this improvement that you have, that your patients are working again, we are walking down five miles rather than one mile? Yeah? If you had a note, the outcome of that, then the Germans will say, you don't have proven, it pays you relevant endpoints, that we need your treatment. So, if you're seeing this plentora

of different systems, it's an ISMS need to understand this carefully, if you advise them, that you are giving pushback very carefully on what they are willing to accept as data set in the Calico trials. From a regulatory perspective, they are trying to look at that minimum. Yeah? The minimum, not because they don't have money, that's one element, but the minimum is to avoid heterogeneity, to avoid the fact that if they are testing five endpoints, that two of them may be negative, yeah? So they're trying to select this one, that certainly will show them a P value. So this is elements that is coming to the table, where you have a conflict between a pure regulatory perspective, and HGA perspective, yeah? And if you have, let's say, a good mix of the two, then you should be designing a strategy of an individual case in Europe, yeah? As as you can, not write a French dossier, even if you are perfect and French, really perfect, yeah? It's not the same as somebody who has a tendon, serve the university in France, in Paris, in this, in the Simone, of Paris, and has, No certain people that are now on the ministry at that in that position. It's so important that somebody is guiding the process for implementation perspective, but as IMS, you're preparing everything so that somebody can do this high level implementation. It's not a translation, but it's a translation of thinking, not of words, not of language, but of thinking, that you need. The same in German, the German perspective, yeah? I mean, even if now at Equic, which is the review board in Germany, most of them are speaking, bluey English. They are not willing to accept English summaries and data sets, but they are willing to accept all the supporting evidence, even if it is in English, yeah? So, you will need to have somebody who is bringing that so called international data set. And, for instance, if you have a value tool kit, view, somebody needs to translate that in the animal system, yeah, that is spitting very carefully. So from that perspective, you can still be successful. In Europe, we have a general data set, but you will have to understand the individual countries, bigger bigularities. And the French are not willing to surrender their system. They were surrendering the regulatory review, yeah? Reluctantly, yeah? So that now somebody in Europe will decide for them, for the French patient. But they have now, let's say, not improve, but they have put a more regular system, even to do the clinical benefit analysis, and a benefit analysis from a market access perspective. So that's their French review that they will be doing. The same for Germany, the same for the UK. The rest of the countries in Europe are following, most of the time, the French system, when it comes to the data set analysis, but when it comes to the negotiational price, they will really look at what price has been approved in other

countries, and you will argue with the same arguments to get that price. And then, present in Spain, the national government will say, Okay, we have an approved product. It has been reviewed, this is the price. And then, the autonomous region of Catalonia, we start like the UK. They will say we will do an HD assessment to see whether this product in Catalonia is cost effective, and there plays very much the core model that is being developed, because you will be able to use your data set for a core model, to implement in countries like Spain, Italy, Sweden, the Netherlands, Norway, the countries that are really doing a cost effectiveness data. We have a globalized cost effect of this model, put in some of the local cost data, not only the US data, but, hopefully, some European data, and if you have some global data, it always helps, actually, to have your model being accepted. And then you can go to market with very strong data. If a product in Europe, it's going to market as a positive review, commercially, this can be a very attractive market, but you're having a lot of hurdles in implementation from an HDA perspective. And that's something that, especially American companies, have difficulties to go with, yeah? They think, oh, here, our big marketing machine in the US, we are preparing our file, we submitting it, and we have a group of, let's say, 10 people that will visit the peers, to inform them about this new product, and why they should be using this product of the pulmonary, and that will be in it. That's not Europe, yeah? I mean, Europe is so much more complicated. You have so many more angles to view that you actually need a very good implementation plan from a European perspective. How you bring this to the table of a potential client is by just saying you need to have a core set of data. Yeah? A core set of data. First degree all, what is your core value that you are bringing to table? Yeah? What is your place in therapy? Have you got data to prove that this place in therapy is valid? And there the, uh, the internal experts are important, but it has to be validated with their external experts panel, yeah? Sometimes the internal battle can be more conservative than the external pattern. So, the advice always should be, bring our value messaging that we are preparing to a review group of experts, yeah, and allow us to present it to them what we think is the true value from a clinical and from a patient perspective, and can you... approve, or can you stop us? With this avenue, this is the type of early validation that will be very important to communicate also to clients, especially if they're a startup client, a startup group, because they have a preconceived idea, sir. The managing director, uh, that has certain ideas, because he spoke to his friend, who is at a clinic in Barcelona, has a peculiar idea, and if that is the case, even if you're doing all this, yeah, I mean, you

will need to have, let's say, a sounding bond to communicate and also your perspective. So that's Europe, yeah? It's a challenging environment. It's a peculiar environment, but it's still a very, very big market, because in all the markets where we have a significant opportunity, where we have a review here, if your product is being funded, you can have access, and if you have access, then patient can start immediately with treatment. A poster for a moment. questions, remarks. And I have lots of questions, but I mean, there's one for sure that I want to ask, but I don't know if anyone else has any burning questions they had in mind. Those from my side, uh, very thanks, was really ensignment. Simon, David, any questions from your guys' side? Any immediate ones that come to mind? Not immediately, actually. And still processing, actually. David, anything from your side? Yeah, I think similar from my side, like a, like a Saimon? I just, yeah, processing so much more, so much, you know, insightful, you know, information, but I just quickly, um, reflect on some, like, recent exercise in the, in the real world world. So for them, for like a, um, whole, uh, you know, very detailed way about like a German market or HTA assessment, you know, uh, procedure, especially how to determine the appropriate, uh, compared to that is exactly what we are working on for one of the recent client work. And another example that are relevant to the card work is properly the societal perspective, uh, where, uh, yeah, like uh, that is what we also exactly, you know, developed it for a C model so far, for uh, genetic therapy. So very, you know, for me, very rather, just one, uh, like feedback is very relevant to my cardboard. Yeah, I want to know more. Yeah, it's very great. Yeah. Thank you. Okay, maybe I'll take us out with some questions and one remark. So I think one remark that I've found to be kind of interesting or at least instructive in sharing with startups is kind of the of Europe, not being scary, but being challenging. I think Paul is Bluebird Bio. So Bluebird Bio, who we talked to, Regetics Bio about, who is our client for many years now, had gene therapy in, um, XVVO gene therapy in, um, Beta Thalosemia, and they're a US-based company, and kind of, like you were saying, Paul. I think thought like a US-based company. You only had US-based employees and then said, yeah, we're going to come into Europe. And they had major issues, like massive issues with HTA. And ultimately, their decision was to completely pull out of Europe. So they went back to the US and they had no sales, no operations at all in Europe. And that was just because of the challenges that they had with HTA. And I think the main issue was kind of to the points you already raised. They just didn't think about it. They just said, we're going to be focussed on the US, we're Americans. We're gonna build an American

focus data set. and value communication, sort of tool and team and everything. And then they must have just decided after the fact, either through whatever investor pressure or just someone having an idea, yeah, let's go to Europe. And okay, they set up some operations there and started engaging with HTAs, but it became pretty evident that it wasn't going to fly for the UK, for Germany, for France, and they basically got shut down everywhere and then announced publicly that they were pulling out of Europe completely and removed all their operations. And I think they've had subsequent issues. And I believe they've sold some of their assets and I don't think are really around as the company any longer. So it's sort of a, an interesting tale that you can share with clients because it shows some of the challenges and the foresight needed. And you can't just act like it's the US and come into Europe. Yeah. So, I mean, the biggest learning from this exercise is that maybe you can use the same data set, yeah, from a clinical perspective, I mean, it's always good to have clinical data in Europe, yeah, if you're coming to Europe. But if you are able to prove to the French that 70% of your patients work Caucasians and your American data centre will accept your data, yeah? If it would be 80 or 90% Hispanics, yeah, and Haitians or whatever, they would say, no, this is not completely our population in Europe, you would like to see, data set in Europe and European patient pool, yeah? But if that, well, it's not the issue. They could manage in the clinical trial design. much better. They're SAP, yeah? Well, what data set, have they pre-specified and how have they analyzed it? And what? I mean, what is the prime the outcome? If you have a claim for long term treatment effect? And your primary endpoint is at 3 months, yeah? show me the data after 3 years or 5 years, huh? How do you collect this data? Although we're not doing that, we're only following up for from a covigilant perspective. That's not good enough in Europe, yeah? I mean, devil need you have. I follow up on this page for longer period of time. Not only for collecting risk benefit factors, but for collecting, maintenance of treatment effect. Yeah, even if this is not anymore, they were blind. This is long term follow up, open label data, it's still supporting evidence that you can present in Europe. But if you don't have that, then we say, I mean, I can only approve your profit for a treatment length of 5 or 6 months, and that's all. Without any chronic or long term treatment effect overall. And if you're doing that, then, going for \$10,000 for just 6 months of alleviation of symptoms, that's not possible to grant this type of price points, yeah. So that's definitely an element that needs can be crafted, yeah. Also... I mean, when we started working with registics bio in 2018, 19, yeah? They were very

suspicious about economic models, yeah? Very suspicious, say, oh, I don't know, like models, we don't know, we don't know. We are now six, seven years later. And, uh, on paper are almost experts in economic model, yeah? They want to be, actually, they will say about it. They want to tell us what ice reacts over there. They are coming from the economic bowl outcome, yeah? So it is possible to change the mindset also of the American companies. But if you are too late in the process, yeah? And the data set is closed, the endpoints are caused this stone, I mean, then, of course, it's an uphill battle that you have. The other question I had was kind of following from my David chair, and so right now, David, myself, and Charles are starting up a project with our Gen X. Um, and it, it sort of is an epi model in a way, uh, which is, is kind of the, I would say, the easy part in one sense. But the origins of the request actually came from Germany and came from, what is the appropriate comparator for, so that this is further MG product VivCard. And the situation there is that it has a specific surological subtype. So like 70% was what it was originally approved for. Now they're doing a label expansion for that other 30, let's say 30%, but they sort of want to know within that 30%, what is the serial of the subtypes within that. Is it 5%, once you're a logical type, you know, 8%, another, 20%, the other or whatever? And then their question is about comparators. So we're actively working on finalizing an SOW on this topic, because it includes a broader abbey model. But hearing you talk about expectations for Germans, I'm kind of curious, like, so the origins of that project, like I mentioned, we're German-based, but I think their ideas like, oh, the, you know, Haas will ask the same question. or similar questions in the UK, there'll be similar questions in the Netherlands and Scandinavia, et cetera, et cetera. So it's now, um, not focussed only on Germany. That was the sort of initiating question, but it's global in nature. But I'm kind of curious what your thoughts are around, like, how the Germans might reflect if we were to stay not have a German data. 'Cause we're just doing a TLR. We were just finding one database, is we're not doing dedicated analyses of a German database, but if we were to find, you know, Dutch data, for instance, or French data, I think that's kind of their ideas, just kind of put the puzzle together and see what data is out there, but I'm kind of curious what your thoughts might be with respect to Eggwig, and would they just shred the, the non-driven data? or... Ignorant will do a literature search. They will do a data analysis of the data sets they have. Yeah. If there are no German data. Yeah. They are willing to accept other European data, and they have a hierarchy. So these are Dutch data, Swedish data, region data, North Europe. Northern Europe is

very believable. Yeah. If you would come with data sets from Nepali, yeah. Yeah, it's more difficult for them, because first of all, they are not able to verify the ability of the data. I'm not saying that they would not recognize the expertise of certain highly reputed Italian professors, but they may have questions about the validity of the database behind the data set, yeah. Is this really validated, yeah? Is this something that can be replicated? If it is, it can be replicated, then it could be relevant, also from a German perspective, but the best is, of course, to go up with Germany, and, you know, German data, Dutch data, even if you're grouped them together, yeah, in order to enrich your data set, and do the analysis. I mean, there, there, of course, you have to prove that you have a homogeneous population that you're doing in your analysis. That's your problem. That will be the question that we... That you have to address... That's a bit, they will say, oh, it's a genius, we can, we can make no valid analysis, so we don't know, and that although they, they cannot give you an added benefit that you have. So if you also not combine every findings about the treatment exposure together. Yeah, I mean, I would be very nervous about that for the reasons of all that. a rare disease too, right? So I think you're going to have big issues with heterogeneity and it'll be almost impossible, I think, to verify like that data set in Sweden of 15 patients is like it could be combined in a reliable, valid way with another data set of 25 patients from France or something. So it's better to just report to or treat. But interest or did papers than combining 20 of all of them? Yeah, I mean, that's honestly becoming my view on it from an epic standpoint everywhere. that's the issue with that example in DM1 that the, uh, that sort of avidity did. They combined a bunch of stuff that shouldn't have been combined for reliability and heterogeneous reasons, but they did it because they wanted a bigger pool. statistically in that sense, but it makes no sense if you just think about it from a reliability and validity standpoint. So... So the French will say we don't know. Yeah? They will give you no rating. The Jervis will say, we don't know, we give you no rating, no added benefit, they will say. Yeah? Don't prove it, no proven as a balance. That's the point. I'm just saying, there's no other. They said, don't prove it at the time. Mm. Yeah. Any other questions, you know, on your side? And this is it also on the German, like the the equic that, Depending on like the clinical size, whether it's like, I think it was like 5% German population, the price negotiation also depend on this, is it less or is it more than 5% of the clinical trial, isn't it? Yeah, I mean... The point in Germany is... The price is a negotiation, yeah? So, in Greek is no negotiation. I Greek is an assessment, yeah? You're not negotiating with equity. The input of equity is

brought to the assessment board, the GBA assessment board. There, we have experts, where there is sickness, one, the ground, and they will look at the data that Equick is presenting. They can overrule Equick. They can say, Ah, there's no German patient in this, sufficiently to be found that we cannot support, let's say, that the findings that you have in your dossier, that equid has reviewed, is relevant for our German patient. That is perfectly possible, yeah? They have the freedom to do that. Also, they have the freedom to say to the assessment of equid, use the equid is negative, and say, we can not assess the data set, because of heterogeneity, but they have a very strong clinical expert at the review boards at GBA, and this expert can say, This is absolutely wrong from a German perspective, because we know that we can rely on international data, for that and that reason, for that, of that, and then you can overturn, overrule the quick assessment. But before you are banking on. Let's have an expert to overrule equic. I mean, this happens only in one other, 20, 30, 40 cases, eh? So this is not a rule. The rule is that equic, the more positive that the equic assessment, this, the easier as mood, the process will be, better by the renew board. And equick, they are not unreasonable. If you are thinking like an equick reviewer, you are presenting the data like an equick reviewer. Yeah. And you can bring in your data set, also uncertain DSA, yes, because of the ultra rare disease state, because of that, yeah? You cannot try to find P value for that and that endpoint, yeah? It's observational, but it gives a dread, yeah. There are elements to discuss, yeah? If it is all princes, cholesterol lowering, yeah? They need to have P values on all the elements, yeah? In order to be sure that it is working, because there are such a huge population, that you can derive your data from. So it has to be extremely strong. But if you have rare and ultra rare diseases, then you can reflect on that. But you have to realize if you have a real military rare disease with extreme small sales potential. You don't have to go through the process. The cutoff point is 25 million euros, yeah? So if you are thinking that you can start with that process and clean this ultra rare, and it will never be able to sell them, then you come to the market with your price without going through the approval. But, of course, companies are very ambitious. They want to say, this patient deserves this treatment, you know, and they will be significantly above the 25 million. And then as soon as it's triggered... And that triggers every view. Yeah, that then you're back to square one, even worse probably. Yeah, that's even worse. Yeah, it was 9 years old. Yeah, yeah. And then you will have to, to either decide later on, either we reduce our price significantly or pull up to, yeah, put up to the market. Yeah, that's that's very

interesting. 25,000 million is a budget impact. It is the present impact for the sickness fund for the gust, no? Okay. Okay. And maybe another question on my side. So now the GCA assessments are running, how do you actually expect, like, would they change the French and the German process at all? Will they? No crystal ball, yeah? The only crystal ball I have is that, um, I never have seen a willingness to surrender their review process. from a French perspective. I was very surprised that the Germans had introduced it, and they were even playing at the introduction of the system, that there would be the most strict of the reviewers of all Europe, yeah? But they were the ones that were not reviewing. They moved from north to the most strict in Europe. So there's almost a competition between the evidence review groups in the UK, French House, Transparency Committee, and Equig, to be the more strict they are. I mean, the highest their reputation is. So, if an assessment will be done from a European perspective, you have to bank on a little cocktail of all these, yeah? And the French will accept a European review, if it is including the French system, plus some of the other European systems, yeah? That will be the minimum for them to agree with that. For instance, you have in Europe, a consultation process where you can have a joint regulatory HDA consultation. Yeah. This is a waste of money to do that, to be honest, huh? I mean, from a clinical perspective, you know what you have to do. If you go there with a HDA. uh, perspective, they will just repeat what I have said, yeah, but they will say, but much clearer, but simpler words they say, stick to our system. If you follow our system, then you have immediately approval, yeah? And they will repeat what are the endpoints, eh? Peace specified, power, no secondary endpoints, yeah? Survival. If you want to claim survival, log down, volleyball, huh? Progressively survival. We doubt it, yeah? We don't believe in patient reported outcomes, so we have some subsidies of that also, but they don't believe in patient reported outcomes, because they are thinking from the bull died professor. Yeah, for them, patients, yeah, it treats patients, but they have no say, yeah? So how can a patient actually decide on a treatment that it is better than something else? Yeah? I mean, the patients are undergoing the treatment. yeah? That's, of course, very different in, in, uh, Anglo-Saxon environment, where patients are much more saying on the table, but not in this type of vibe. So you have to understand without the sensitivities of the reviewers. in order to present and push the data. No, indeed. And, uh, we have about five minutes left. I had one last question. Well, and anybody's kind of touched on it before, but I'm sort of curious in this context. So, in the example of our text, for instance, one of the things that be included in the

proposal, going to the point that you raised is, like, validating some of the value messaging or value story, in essence, and, you know, given what you had shared, it obviously suggests it's important to do that in a couple different markets, because you're likely gonna get different feedback. But what would you say in terms of like the, is there an ideal number of pairs that you would say? Because I think in this artech situation, we saw the other competitor of ours had like 50 reviews or something that are like, you know, interviews they were doing with pairs, which is seen astronomically high and almost like, what are you going to get from that many? But do you have a sense on like where an ideal balance is? Like if you do one in each country, is that sufficient or just you get the land of the land, at least? If you want to do, if you want to have a real input in the clerical trial program, Yeah. From an HDA perspective, you need to have a full former member of the transparency committee approach. Yeah, yeah, yeah. You need to have somebody who has reviewed equic data. Yeah? Or at least has been sitting on the GBA review board in Germany. So that he knows how the equic data are read by the other members of the review board and see whether this is, okay, from a UK perspective, there's so many consultants there. So you need somebody that has been an expert at the review, uh, both of the nice committee, yeah, expert from a cost-effectiveness perspective, not so much from a clinical perspective, because the only country where you really can focus on cost-effectiveness, it will be the UK, so we need to have one view there. The rest will be clinical and added value. So if you have three very good candidates. Yeah, that will help you a lot. Yeah? If you do interviews with 50, you will have... It's interviews with, uh, things of very important, this type of thing, this is not... You have to go in dialogue with the speed, I say, this is actually what the disease shows us, yeah? What can we do? If we impact here, what will that be? And this gives the opportunity to discuss. And they will say, yes, we have had a different, a similar situation. We have a different disease, but a similar challenge, and this was addressed appropriately, or not appropriately, because of that, and that reason, that, then you learn a lot more than trying to say is this sufficient, no, or yes, type of thing. Yeah, condition. You have to come into contact with his curses. No, yeah, that's another, I don't know. We go during the UK, we were in contact with someone, but for the remaining... Yeah, I had some ideas there, but that's maybe not for this discussion necessarily, but I think it kind of reiterates the importance just of the fact knowing that how different they review the data and even, you know, it could be identical data that they perceive differently. So I think it sort of reiterates the importance of having

some discussions with them at a stage where you can still impact something. Okay. And also what you have to understand is, um, Some companies are moving into the so-called, not uh, doctor focussed, but patient focussed. Yeah? So, patient reported outcomes. A strong focus, we had a GNG also, we had a group, we were into PROs, and every research project needed to have a PRO, questionnaire attached to it, support. You have prepared a lot of slides of PROs. I can no more agree with the fact that if you have a PRO, It's icing on the cake, yeah? But if there is no cake, yeah? There is no need for icing, yeah? So it's better to have a very solid cake and no icing. Then having an average cake, yeah, very ill baked. We have a lot of icing on top of that, because they will not eat the icing at all. They will say, I mean, you have no problem, anything, from a clinical perspective, yeah. How can you now try to make, so called, through the eyes of the patient, claims that your product is needed? Yeah. So, there's a lot of talking in the industry that patient and face report doubles are very important, and they will drive the value. This is not a European reality, yeah? I mean, in all mission statement, they the same patients are coming first, and we need to have the best help for our patient population and so forth, and they are the ones, but they will never allow patient to decide, uh, I like this product more than the other one, yeah, because it gives me a better outcome. Yeah? It will not drive the selection of the technology and certainly will have no positive input or impact on price potential. Yeah. Yeah. And sometimes, if you have a discussion, 50% of the times they are discussing PROs, yeah? Because it's so, and sometimes you need to be a road to determine utilities, yeah. But even then the discussion is, if this is collected in a clinical trial, sponsored by a pharmaceutical company, shall we consider it, yeah? It's coming from an academic institute that has no affiliation with a sponsor, then the willingness to accept the data, from an HDM perspective, are significantly higher, you know? I mean, we are suspicious, right? I mean, they're looking at us as strongly biased and treating a Spanish suspiciously, almost like, a little bit like, specimen, because, I mean, the industry and the companies that are working for the industry. So this was a short introduction to HD at Europe. I mean, whatever we want to discuss some specific countries, there's always an opportunity to go in a much more detailed, yeah? But at least, this gives a helicopter.